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Spectrogram-Based Transfer Learning for Electrocardiogram (ECG) Artifact Detection in the ICU

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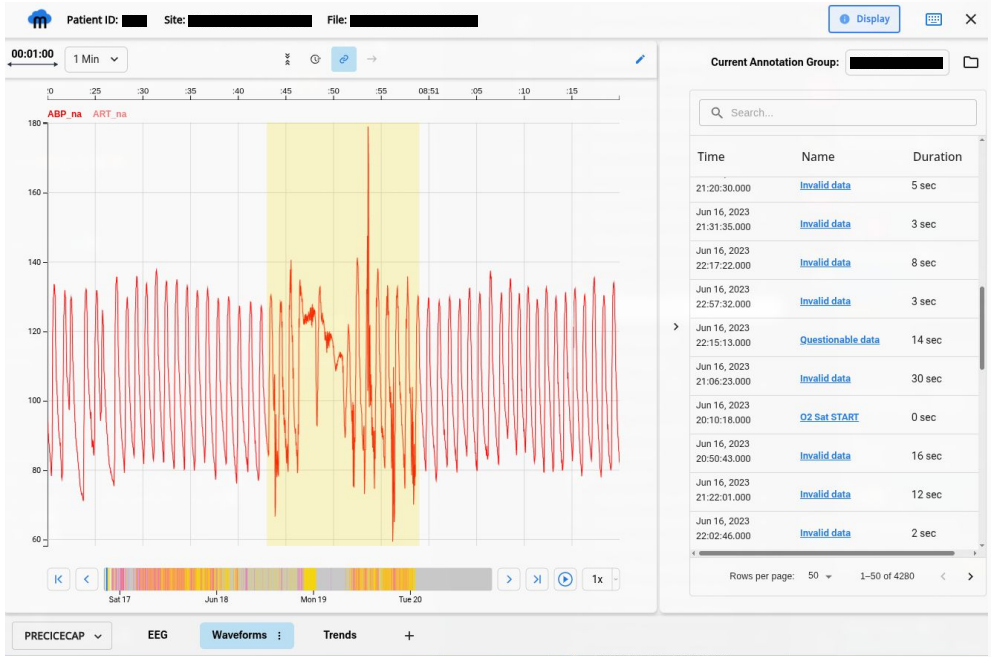
Disclosures

- Tony K. Okeke, Ethan Moyer, and Richard Moberg are employees of Moberg Analytics, Inc.
- All other authors report no relevant financial relationships.

The Problem: ECG Artifact in the ICU

- Continuous ECG monitoring is critical but frequently degraded
- Sources: patient movement, muscle contraction, electrode issues, electrical interference
- Impairs clinical interpretation and downstream ml/research applications



Why This Matters

- ~30% of ICU monitoring time affected by artifact
- 6,963 hours monitored → ~2,100 hours potentially corrupted
- Current approaches: Statistical Signal Quality Indices (SSQIs)
  - SSQIs use fixed thresholds, struggle with non-stationary ICU data
- Need: Robust, generalizable, HIGH-SENSITIVITY detection

Our Hypothesis

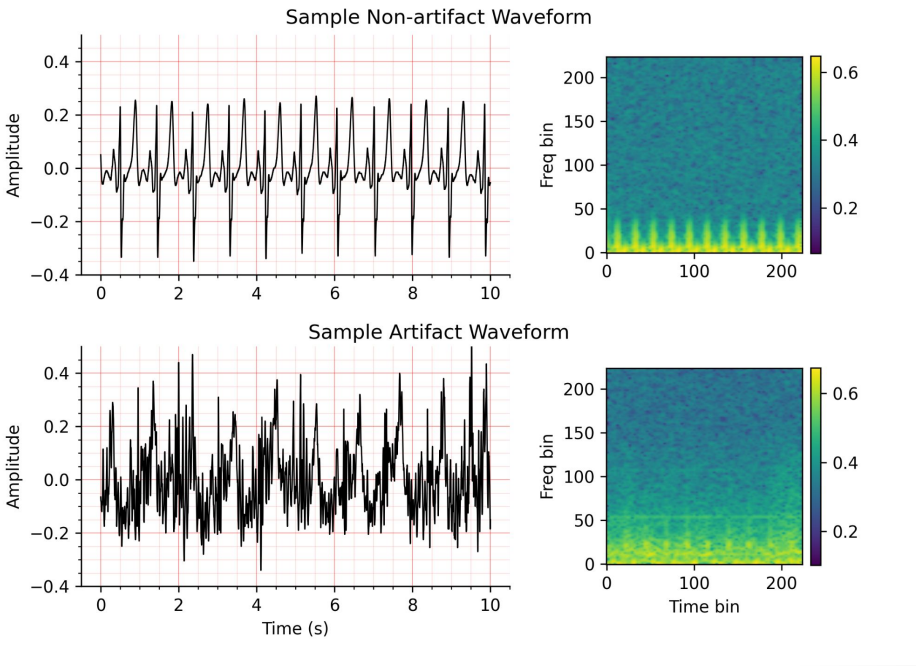
- **Hypothesis:** Transfer learning on ECG spectrograms would outperform SSQI-based approaches
- **Key insight:** Spectrograms capture time–frequency patterns (broadband noise, intermittent interference) that CNNs can learn to discriminate
- **Design priority:** High sensitivity (≥95%) to support downstream data curation

Data & Annotation

- PRECICECAP study: 65 post-cardiac arrest patients, 7 sites
- 10,890 ten-second ECG segments (Lead II, 500 Hz)
- 6,963 hours of continuous monitoring
- Two-stage annotation process:
  - Stage 1: Research assistants flagged artifact regions (consensus)
  - Stage 2: Two ICU physicians scored each segment (4-point scale)
- 19.6% cross-bin disagreement → resolved by adjudication
- Final: Categories 1-2 = Non-artifact, Categories 3-4 = Artifact

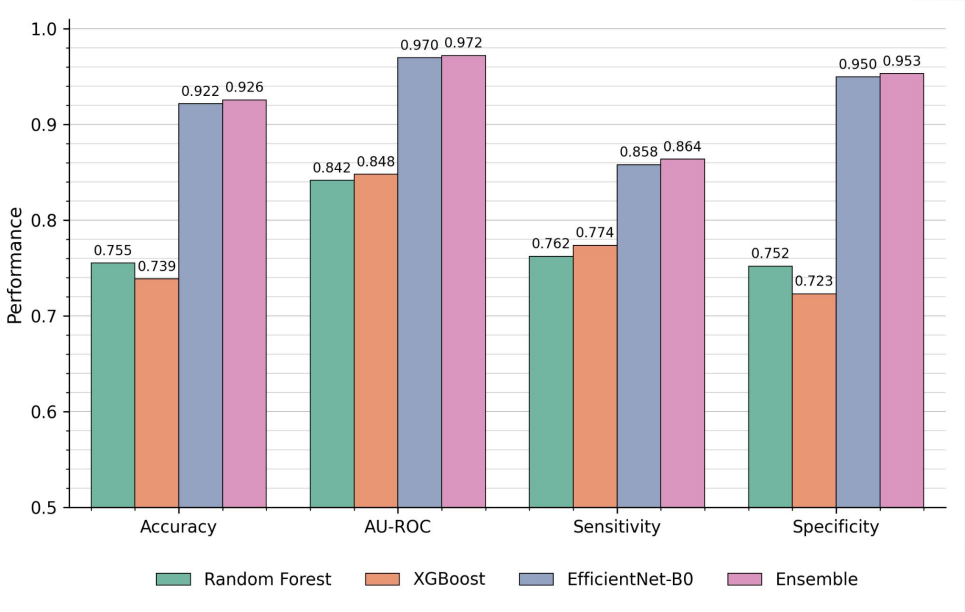
Methods — The Spectrogram Approach

- STFT: FFT size 256, hop 64
- Spectrograms → 224×224 RGB
- EfficientNet-B0 (ImageNet pretrained)
- Full fine-tuning on ECG data
- Baseline: 5 SSQIs → XGBoost



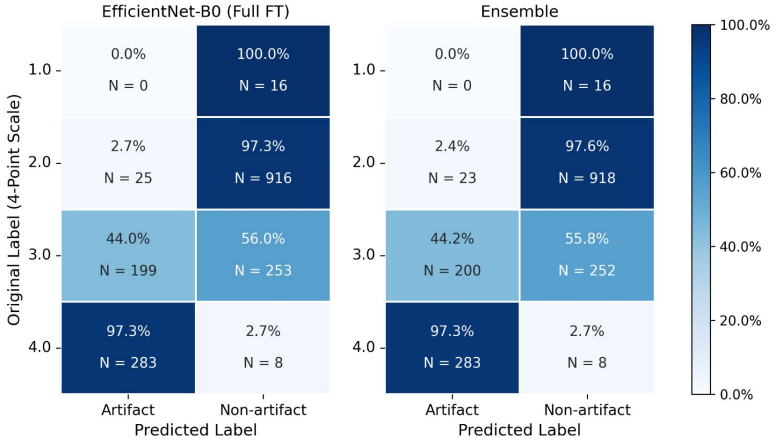
Results — Model performance

- Deep learning substantially outperforms SSQI baselines
- **EfficientNet-B0:** AU-ROC 0.970, accuracy 92.2%, sensitivity 85.8%
- **Best SSQI (XGBoost):** AU-ROC 0.848, accuracy 73.9%
- Ensemble offers marginal gain (0.972) — not worth the complexity



Where Does the Model Fail?

- **Perfect at Extremes:** 100% correct on category 1 (perfect ECG), 97.3% on category 4 (unrecognizable)
- **Struggles at the Boundary:** 56% of category 3 misclassified as non-artifact
- Model fails where humans disagree — a reasonable failure mode



Conclusions & Clinical Implications

- Spectrogram-based transfer learning provides superior artifact detection vs SSQIs
- Enables high-sensitivity curation of continuous icu monitoring data
- Practical recommendation: use EfficientNet-B0 full fine-tuning; recalibrate thresholds for your cohort
- Cross-modality applicability demonstrated (arterial blood pressure)
- Patient-specific fine-tuning can boost specificity with just 20 labeled segments

Limitations & Future Directions

- Limitations:
  - Single cohort (post-cardiac arrest) — generalizability to be determined
  - Downstream impact not yet quantified

- Future work:
  - External validation across populations and equipment
  - Ordinal regression to predict 4-point score directly → risk-based filtering
  - Real-time implementation

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